

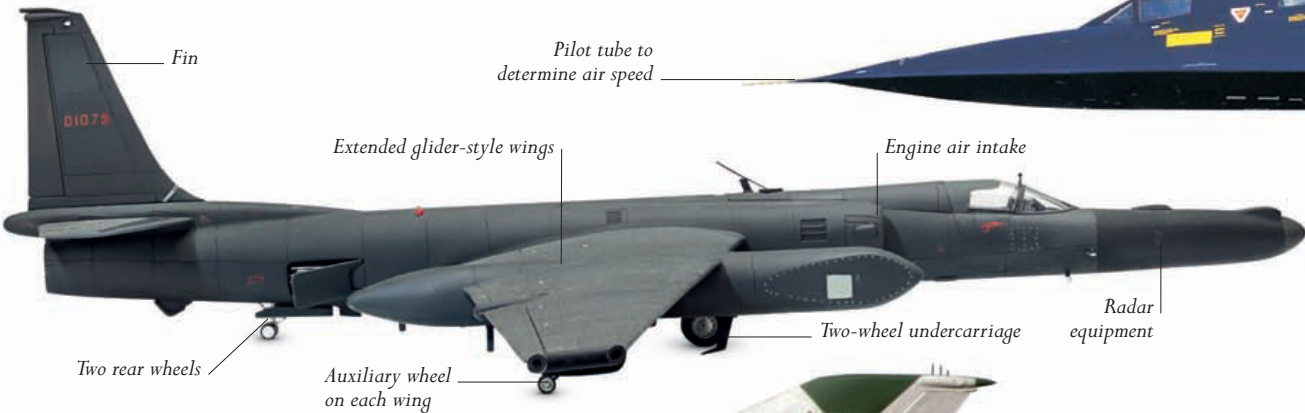
ELECTRONIC WARFARE AND RECONNAISSANCE AIRCRAFT

Aerial reconnaissance plays a vital role in intelligence gathering, whether at a tactical level via the MiG-21 and OV-1 Mohawk, or at a strategic level via the U-2 or SR-71. These “eyes in the sky” reached a new level of sophistication with the introduction of Airborne Early Warning (AEW) aircraft, which can build up an electronic picture of enemy movements from enormous distances. These in turn have evolved into Airborne Warning and Control System (AWACS) aircraft, which provide all-weather command, communications, and surveillance.

► LOCKHEED U-2

Date	1957	Origin	US
Wingspan	31.4m (103ft)		
Length	19.2m (63ft)		
Top speed	805kph (500mph)		
Engine	8,618kg (19,000lb)	General Electric F118-101 turbofan	

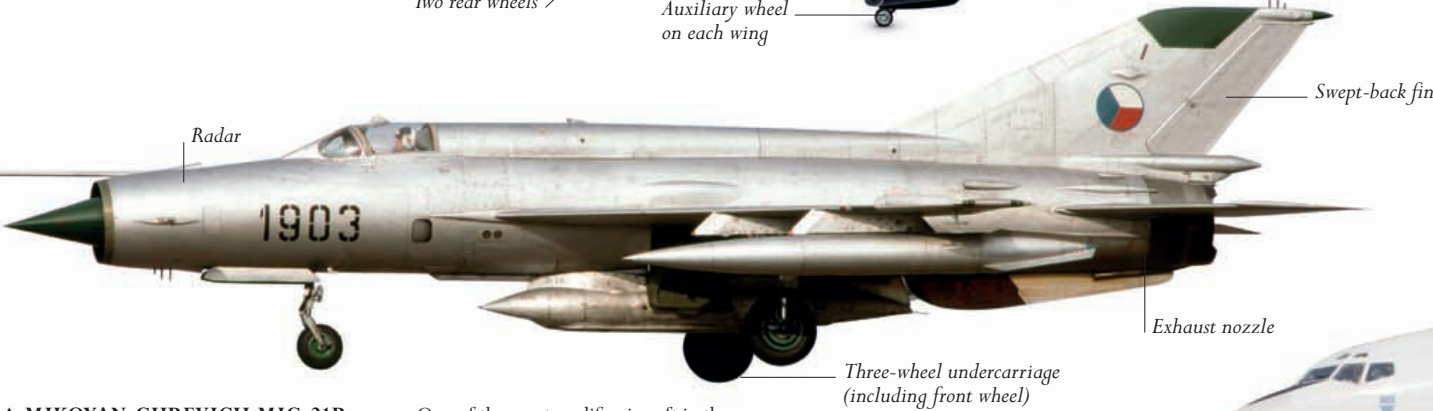
A strategic reconnaissance aircraft, the U-2 – capable of cruising at heights of 21,000m (70,000ft) – was used by the Central Intelligence Agency (CIA) as a spy plane. It earned infamy when one was shot down in Soviet airspace in 1960.



▲ MIKOYAN-GUREVICH MIG-21R

Date	1959	Origin	Soviet Union
Wingspan	7.15m (23½ft)		
Length	15.76m (51¾ft)		
Top speed	2,230kph (1,385mph)		
Engine	6,600kg (14,550lb)	Tumansky R-13-300 turbojet	

One of the most prolific aircraft in the Soviet armoury, the MiG-21R was a tactical reconnaissance variant in which guns were replaced by optical and infrared cameras, either mounted within the airframe or in external wing pods.



Chin bulge with Electronic Surveillance Measures (ESM) suite



◀ GRUMMAN OV-1 MOHAWK

Date	1959	Origin	US
Wingspan	14.63m (48ft)		
Length	12.5m (41ft)		
Top speed	724kph (450mph)		
Engine	Two 1,044kW (1,400hp)	Lycoming T53-L-701 turboprops	

Designed for battlefield surveillance, the Mohawk was fitted with optical cameras and a Side-Looking Airborne Radar (SLAR) that provided a detailed picture of enemy movements on the ground.



▲ LOCKHEED P-3C ORION

Date	1962	Origin	US
Wingspan	30.4m (99¾ft)		
Length	35.6m (116¾ft)		
Top speed	750kph (466mph)		
Engine	Four 3,700kW (4,600hp)	Allison T56-A-14 turboprops	

Developed from the Electra commercial airliner, the Orion acts as a maritime surveillance and anti-submarine aircraft. Its Magnetic Anomaly Detector (MAD) boom is used to track the presence of submerged submarines.